

# TrES-3b

R-1639

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-3\_151016 · created 2026-08-02T06:37:32+00:00

## BENCHMARK: MARGINAL

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457311.6128 +/- 0.0011 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.197 +/- 0.021                 |
| Transit depth (Rp/Rs)^2                           | 3.86 +/- 0.83 [%]               |
| Orbital Inclination (inc)                         | 81.34 +/- 0.47                  |
| Airmass coefficient 1 (a1)                        | 1.224 +/- 0.011                 |
| Airmass coefficient 2 (a2)                        | -0.1232 +/- 0.0099              |
| Scatter in the residuals of the lightcurve fit is | 0.9 %                           |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 4.18                            |
| Optimal Annulus                                   | 11.35                           |
| Transit Duration (day)                            | 0.0516 +/- 0.0038               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.197 ± 0.021 vs 0.16309 (deviation 1.45σ) |
| Duration fit vs published | 0.0516 d vs 0.0567 d (deviation 1.2σ)      |
| Mid-time O–C              | -3.8 min                                   |
| Depth SNR                 | 8.4                                        |
| Residual scatter          | 0.9 %                                      |

### Light curve

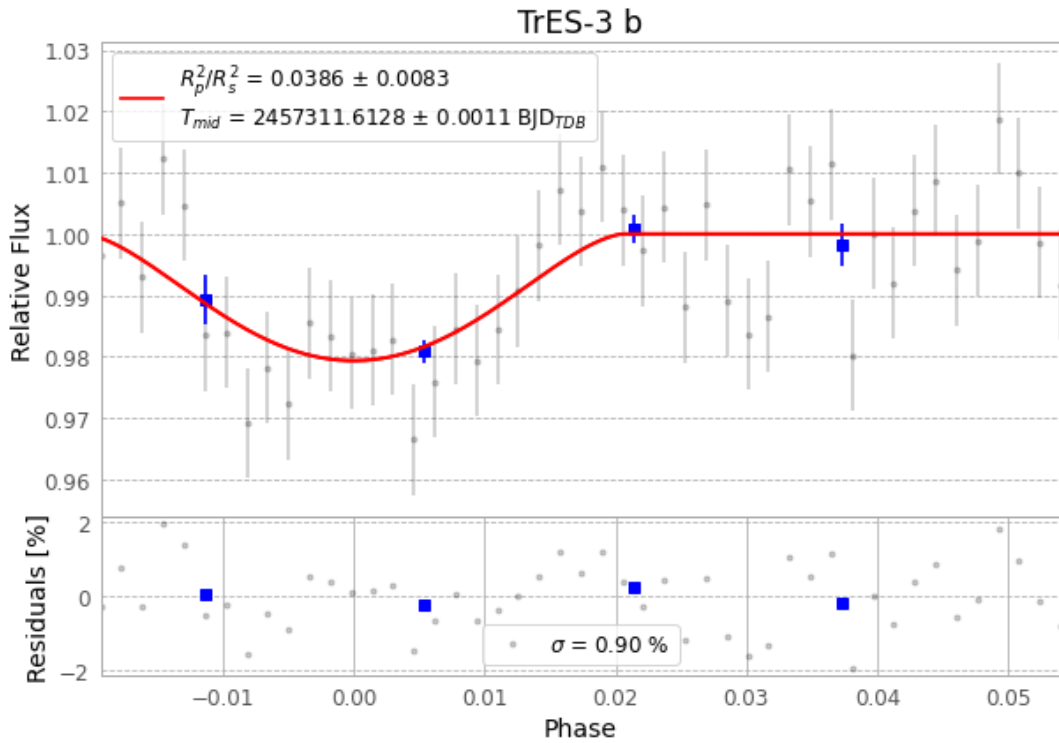


Figure 1 — FinalLightCurve\_TrES-3 b\_2015-10-15.png (SHA-256 14de2c6fb150c392...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [181, 278], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 648b3972f7fc848a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

47 FITS frames (31 MB), TRES-3151016020615.FITS → TRES-3151016042412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-151016.log (f7d3dabf2fc7...), FinalLightCurve\_TrES-3 b\_2015-10-15.png (14de2c6fb150...), FinalLightCurve\_TrES-3 b\_2015-10-15.csv (cff5882ee52e...)

## Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 06:37Z.